

Divya Natekar

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Education

New York University, Tandon School of Engineering

Brooklyn, NY

Master of Science — Urban Data Science, GPA: 3.61/4.0

Sept 2024 – May 2026

◦ **Courses:** Data Analysis, Machine Learning, Virtual & Augmented Reality, Artificial Intelligence, Advanced Spatial Analytics, Urban Computing & Artificial Intelligence

Research Interests

Urban Artificial Intelligence; Agent-Based Modeling; Human Mobility Analysis; Spatial Data Science; LiDAR Point Cloud Processing; Explainable Urban Models; Large Language Models & Generative AI Architecture

Research

Thinking on the Move: Agent-Based Urban Simulation Decision Modeling

Sept 2025 – May 2026

NYU Center for Urban Science + Progress & Downtown Brooklyn Partnership

(Project Sponsor: Mark Landolina, Senior Director, Real Estate and Economic Development, Downtown Brooklyn Partnership; Mentor: Prof. Takahiro Yabe, Assistant Professor, Department of Technology Management and Innovation and CUSP, NYU Tandon)

- **Agent-Based Urban Simulation:** Developed an agent-based simulation framework for analyzing urban mobility flows and multi-agent choice behavior in dense metropolitan environments.
- **LLM-Enhanced Choice Modeling:** Integrated large language models, spatial interaction methods, and gravity-based preference engines to model qualitative decision-making beyond simple spatial distance.
- **Urban Data Integration:** Processed high-dimensional mobility datasets, review text embeddings, and spatial POI data to evaluate local movement patterns and human behavior trends.
- **Planning Decision Support:** Built predictive simulation utilities to support stakeholder-oriented urban planning, infrastructure optimizations, and spatial scenario evaluations.

Transient Object Removal from Urban LiDAR Point Clouds

Jun 2025 – Aug 2025

NYU CUSP Summer Guided Research (Mentor: Prof. Debra Laefer)

- **Machine Learning Workflow:** Designed and implemented filtering workflows using RANSAC algorithms and spatial clustering to remove transient urban objects (pedestrians, vehicles) from 3D LiDAR point clouds.
- **3D Urban Modeling:** Improved fidelity, spatial consistency, and geometric accuracy of 3D urban models using spatial filtering and point-cloud processing techniques.
- **Research Recognition:** Selected for the NYU Tandon CUSP Experiential Learning Scholarship for Summer 2025 guided research.
- **Presentation:** Presented research findings at the BATWorks Climate Event and NYU CUSP Fall 2025 Research Showcase.

Experience

PM Accelerator

Remote

Software Engineer Intern — AI/ML, Lead Engineer

Aug 2026 – Oct 2026 (Expected)

- **AI Decision Engine & REST APIs:** Designed and deployed Next.js App Router API micro-routes (`/api/people`, `/api/decisions`) backed by Supabase and PostgreSQL to power a real-time AI personal relationship manager.
- **LLM Telemetry & Monitoring:** Integrated Langfuse SDK telemetry across Next.js API boundaries to trace LLM execution flows, evaluate prompt latency, and log real-time inference inputs.
- **Context Pipeline Architecture:** Engineered snapshot transformation utilities mapping asynchronous database state into structured context payloads (`User`, `Person`, `Interaction History`) for prompt evaluation.
- **Database Optimization & Security:** Refactored query logic by implementing ISO timestamp boundary filtering (`.gte/.lte`) in Supabase, replacing regex searches and ensuring data security across server routes.
- **Engineering Leadership:** Led code reviews, resolved Git branch rebasing/merge conflicts, and enforced system standardization across the dev team.

L&T Technology Services (LTTTS)

Airoli, India

Software Development Intern

Jan 2023 – Jul 2023

- **Backend Development:** Built consumer and backend components for a proof-of-concept healthcare data platform integrating Connected Wearable Devices with the Philips HSDP cloud ecosystem.
- **Real-Time Data Engineering:** Developed real-time data ingestion and processing pipelines using Java Spring Boot, Apache Kafka, and Spring Vault.
- **Cloud Infrastructure:** Provisioned and managed cloud infrastructure using Terraform and Microsoft Azure services.
- **DevOps & APIs:** Implemented CI/CD workflows through Azure DevOps, performed API testing with Postman, and documented services using Swagger within Agile development workflows.
- **Data Analysis:** Conducted data analysis and visualization workflows in Python to generate insights from healthcare datasets.

Technical Skills

Programming & Languages: Python, TypeScript, Java, C++, SQL, HTML/CSS

AI/ML & Frameworks: Large Language Models (LLMs), Generative AI, PyTorch, TensorFlow, Scikit-learn, Langfuse (LLM Observability), RAG, Prompt Engineering, Computer Vision, Embeddings, 3D LiDAR Processing, RANSAC, Pandas, NumPy

Full-Stack & Web Development: Next.js, React.js, Java Spring Boot, REST APIs, Tailwind CSS, Microservices Architecture
Cloud, Databases & DevOps: Supabase, PostgreSQL, Microsoft Azure, Apache Kafka, Terraform, Docker, Azure DevOps, Git, GitHub, Spring Vault, Swagger, Postman

References

Debra Laefer: Professor at the NYU Tandon School of Engineering and the Center for Urban Science + Progress, New York University.